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Comparison and Combination of Factorial and Monte-Carlo Design in Sensitivity Analysis

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The sensitivity of building physics simulation programs to the uncertainty of input parameters is analysed. A comparison is drawn between the classical Monte-Carlo method and a new method based on fractional factorial design. Their numerical behaviour is analysed and their respective advantages are discussed. The new method provides more information for an equivalent number of simulation runs. It allows a real feedback process since the relation between the uncertainty of the input and output data is fitted on a linear model. It also allows a better understanding of the behaviour of the studied building. Both methods are illustrated with a three-storey building, in which ventilation is analysed for different ratios between the wind and buoyancy forces. However, the two methods can also be combined and a map of the uncertainty of the output for the whole domain, determined by the variation of a few input parameters, can be drawn.

NOMENCLATURE

X_i	input parameter
α_i	coefficient of a model
e_{ij}	value of the parameter X_j during the simulation i
E	matrix of e_{ij}
μ_i	real mean of Y_i
σ_i	real standard deviation of Y_i
R	resolution level of a fractional factorial design
Y_i	output parameter
α	probability of a confidence interval
$m(Y_i)$	estimate of the mean of Y_i after a series of runs
$s(Y_i)$	estimate of the standard deviation of Y_i after a series of runs
$t_\alpha[N]$	student distribution quantile corresponding to N degrees of freedom and α probability
$\chi^2_\alpha[N]$	chi^2 quantile corresponding to N degrees of freedom and α probability
ΔT	indoor–outdoor temperature difference [K]
g	gravitational acceleration [m s^{-2}]
h	warm zone height [m]
T_{in}	indoor temperature [K]
v	wind speed [m s^{-1}]
Q_{ij}	element of the flow matrix [kg s^{-1}]
q_{ij}	mass flow though an element [kg s^{-1}]
C_{ij}	airtightness coefficient of an element [kg s^{-1} Pa^{-n}]
ΔP_{ij}	pressure difference between zones i and j [Pa]
n_{ij}	exponent of the flow equation for an element

INTRODUCTION

THE aim of the sensitivity analysis is to assess the effect of input parameter variations on the output data of a simulation or measurement process. In the validation procedure of a building simulation program, this also corresponds to studying the effect of input uncertainties on the output confidence interval.

The usual way of estimating the sensitivity of a process by simulation is the well-known Monte-Carlo method [1]. We propose another method in this paper: the factorial design, which is less used, but gives more information and thus provides real feedback on the input.

Ideally, the two methods can be combined to estimate the variation of the sensitivity in a multidirectional experimental space.

Both methods are presented here applied to the air infiltration field because this is our area of investigation. Building physics is an area which needs suitable methods for sensitivity analysis because the data used in simulation have large confidence intervals [2].

The confidence intervals of simulation data are required to compare measured and simulated data for empirical validation. In that perspective, it is essential to identify the input parameters which affect the output accuracy. However, in the design process, it is also important to know the minimal required accuracy of the input parameter necessary to warrant a given accuracy of the output. The proposed method is equally interesting for researchers and designers.

The presented results belong to a larger study on the sensitivity of measurements and simulations in the framework of Annex 23 'Multizone Air Flow Modelling' of the International Energy Agency program ECB&C[3].

The ultimate aim of the design of experiment is to obtain the required information with a minimum of work and the highest confidence possible. If the experiment set is not optimised by means of a suitable analysis procedure before the experiment starts, the researcher is later con-

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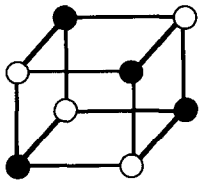


Fig. 1. Space position of a factorial design of dimension 3.

strained to use sophisticated statistic tools to analyse his data, risking at the same time that the interesting information is not in the data obtained. An argument which supports the claim for optimal design is that in any experiment there is a design in the sense that it is always possible to write a matrix with the values of each parameter for each experiment.

The 'star' design which means varying one parameter at a time is not a satisfactory design because no interaction effects can be estimated that way and the distribution of information within the experimental space is not at all uniform.

FACTORIAL DESIGN

Factorial design is a classical experimental design method which allows the determination of the coefficients α_i, α_{ij}, etc. of a linear model with interactions as in equation (1) with a minimum of points.

$$Y = \alpha_0 + \sum_i \alpha_i X_i + \sum_{i \neq j} \alpha_{ij} X_i X_j + \dots \quad (1)$$

The matrix of experiments E includes the elements e_{ij} which are the values of the input parameters X_j for the experiment i. In factorial design, measurements or simulations are performed only at the minimum and maximum values of each of the N parameters. As all the combinations are used, 2^N points are defined in the experimental space, corresponding to the vertex of a parallelepiped of dimension N.

If the input parameters are centred and normalised, the matrix of experiments can be written with -1 and $+1$ values. This gives, in the case of three dimensions, the matrix presented in equation (2) and illustrated in Fig. 1.

Factorial design allows the identification of the 2^N coefficients of a saturated model such as that of equation (1). If not all the coefficients are interesting, but for example only the first orders of interaction are required, it is possible to use a fraction of the full factorial matrix. The fractional matrix is built from one or more relations called generators, which define which coefficients are aliased.

Without entering into too much detail (interested readers will find more information in [4]), we can say that fractional factorial designs are sorted according to their resolution R. The resolution index indicates which kind of coefficients are confounded, as explained in Table 1.

At present, there are no straightforward algorithms to build the suitable fractional factorial design from the desired coefficients. It is necessary to use tables (cf. [4]) or a trial and error method to obtain satisfactory matrices.

$$E = \begin{bmatrix} -1 & -1 & -1 \\ -1 & -1 & +1 \\ -1 & +1 & -1 \\ -1 & +1 & +1 \\ +1 & -1 & -1 \\ +1 & -1 & +1 \\ +1 & +1 & -1 \\ +1 & +1 & +1 \end{bmatrix}. \quad (2)$$

MONTE-CARLO DESIGN

In the Monte-Carlo design, the points of simulation in the experimental space are chosen at random [5]. A probability distribution is associated with each input parameter, representing the quality of the information available to the modeller. The simulation points are chosen at random with respect to these distributions. The output data obtained after each run are saved and then analysed to extract statistical information, allowing the qualification of the statistical behaviour of the output parameters Y_i. One is usually interested in the probability distribution of Y_i, their mean values $m(Y_i)$ and their standard deviations $s(Y_i)$.

Estimates of the mean and of the standard deviation of the population of the output parameters can be calculated after any simulation number (>1), but it is the only information available from this method: a mean answer and its standard deviation, related to the (input) experimental domain. This space is determined by the combination of the definition set of input probability distributions.

For a large number of input parameters, whatever their probability distributions, one expects answers to be more or less normally distributed if the simulation process is continuous. The estimates of the confidence intervals of the mean $m(Y_i)$ and the standard deviation $s(Y_i)$ are based on the hypothesis that the output distributions are Gaussian.

The confidence interval of the real mean μ_j of the output Y_i follows a Student law T with $N-1$ degrees of freedom. That means that if γ is the confidence level of a two-sided test, usually 95 or 99%, the real mean μ_j verifies the probability equation

$$\text{Prob}[(T: N-1) \leqslant \mu_j] = \tfrac{1}{2}(1+\gamma). \quad (3)$$

From the tabulated values $t_{1-\gamma/2}[N-1]$, the confidence interval of μ_j is then given by

$$m(Y_j) - \frac{s(Y_j)t_{1-\gamma/2}[N-1]}{\sqrt{N}} \leqslant \mu_j \leqslant m(Y_j) + \frac{s(Y_j)t_{1-\gamma/2}[N-1]}{\sqrt{N}}. \quad (4)$$

The confidence interval of the standard deviation σ_j^2 follows a χ^2 law with $N-1$ degrees of freedom. That means that if γ is the confidence level of a two-sided test, the real standard deviation σ_j^2 verifies the probability equations

$$\text{Prob}[(\chi^2: N-1) < \sigma_j^2] = \tfrac{1}{2}(1+\gamma) \quad (5)$$

Table 1. Resolution of fractional factorial matrices

Resolution	Definition
R = III	does not confound main effects with one another but does confound main effects with two-factor interactions
R = IV	does not confound main effects and two-factor interactions but does confound two-factor interactions with other two-factor interactions
R = V	does not confound main effects and two-factor interactions with each other, but does confound two-factor interactions with three-factor interactions

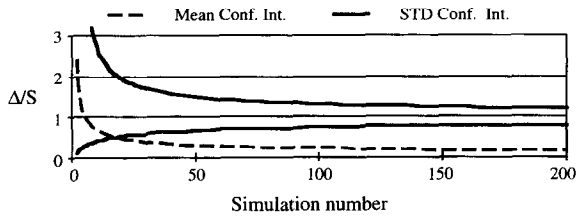


Fig. 2. Relation between the number of runs and the 95% confidence intervals, normalised by the standard deviation $s(Y_j)$.

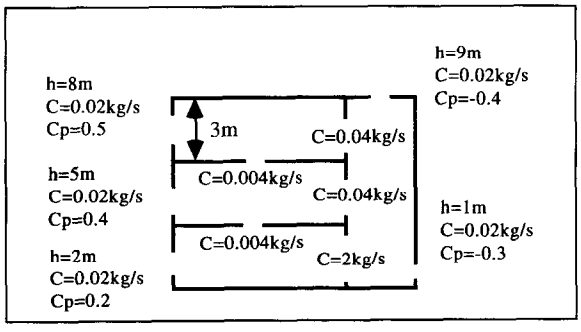


Fig. 3. The paper building USERTEST1.

$$\text{Prob}[(\chi^2 : N-1) > \sigma_j^2] = \tfrac{1}{2}(1-\gamma). \tag{6}$$

From the tabulated values of $\chi^2_{\gamma/2}[N-1]$ and $\chi^2_{1-\gamma/2}[N-1]$, the confidence interval of σ_j is then given by

$$(N-1)\frac{s(Y_j)^2}{\chi^2_{\gamma/2}[N-1]} \leqslant \sigma_j^2 \leqslant (N-1)\frac{s(Y_j)^2}{\chi^2_{1-\gamma/2}[N-1]}. \tag{7}$$

Figure 2 shows the relation between the number of runs and these confidence intervals normalised by the standard deviation $s(Y_j)$. The boundaries of the confidence interval of the standard deviation tend towards 1, however after 60–80 runs, further improvement of the accuracy is very small. This behaviour is independent of the input parameter number and makes the accuracy of the Monte-Carlo method wholly dependent on the number of runs.

CASE STUDY

In order to test the respective advantages of both methods presented above, a building has been chosen among the test cases used within the IEA ECB Annex 23 'Multizone Air Infiltration Modelling'. The building corresponds to a two-dimensional network with three vertically superposed zones and a stairwell on the three floors. The characteristics are presented in Fig. 3 and Table 2. All the outdoor–indoor conductances have the same airtightness. Between the zones and the stairwell, the conductances modelling the door are less tight, especially on the first floor. The leakages through the ceilings are small. The tested parameters are listed in Table 3.

Table 2. Characteristics of the paper building USERTEST1

Volumes	stairwell	135 [m^3]
	each floor	150 [m^3]
	total	585 [m^3]
Airtightness	n_{L50}	6 [ach]
Flow exponent	n	0.66

MODEL COMVEN

COMVEN is a code developed by the COMIS research group for air flow and pollutant transport simulation in multizone buildings; the structure of the building is modelled as a network in which the nodes correspond to the zones and the edges correspond to the air flow patterns between zones.

The scientific basis of the program is described in [6]. Numerically, the program is based on the Newton–Raphson method which integrates the differential system of equations [7]. The flow is modelled by a power law

$$Q_{ij} = C_{ij}(\Delta P_{ij})^{n_{ij}}. \tag{8}$$

ARCHIMEDES NUMBER

The Archimedes number Ar is defined as

$$Ar = \frac{\Delta T g h}{T_{\text{in}} v^2}. \tag{9}$$

This number corresponds to the ratio between stack and wind induced forces. Choosing a single indoor temperature can be simplistic in a typically multizone case but allows the definition of a reference to the meteorological situation to compare the behaviour of different buildings.

TREATMENT WITH FRACTIONAL FACTORIAL DESIGN

Design

A $2_{\text{IV}}^{(24-16)}$ fractional factorial design was used. It allows in 256 runs the determination of 136 coefficients among the 301 ones corresponding to a linear model of 24 parameters. It is a design of resolution IV, which means that the main effects α_i are neither aliased between themselves, nor with first order interaction coefficients, whereas the

Table 3. Tested parameters

	Description	COMVEN parameters
1.	Elementary indoor–outdoor airtightness	CR_OUT
2.	Indoor–outdoor exponent	n_OUT
3.	Airtightness between the floors	CR_FL
4.	Exponent between the floors	n_FL
5.	Airtightness between zone 1 and stairwell	CR_ST1
6.	Exponent between zone 1 and the stairwell	n_ST1
7.	Airtightness between zone 2 or 3 and the stairwell	CR_ST2
8.	Exponent between zone 2 or 3 and the stairwell	n_ST2
9.	Temperature in the 1st floor	T_fl1
10.	Temperature in the 2nd floor	T_fl2
11.	Temperature in the 3rd floor	T_fl3
12.	Temperature in the stairwell	T_st
13.	Temperature gradient in the stairwell	grad(T)
14.	Windward pressure coeff. in front of zone 1	Cp(1)
15.	Windward pressure coeff. in front of zone 2	Cp(2)
16.	Windward pressure coeff. in front of zone 3	Cp(3)
17.	Pressure coeff. on the roof	Cp(4)
18.	Leeward pressure coeff. at back of zone 4	Cp(5)
19.	Wind profile coefficient	W_profil
20.	Wind speed	W_speed
21.	Outdoor temperature	Tex
22.	Outdoor humidity	Humidity
23.	Atmospheric pressure	Patm

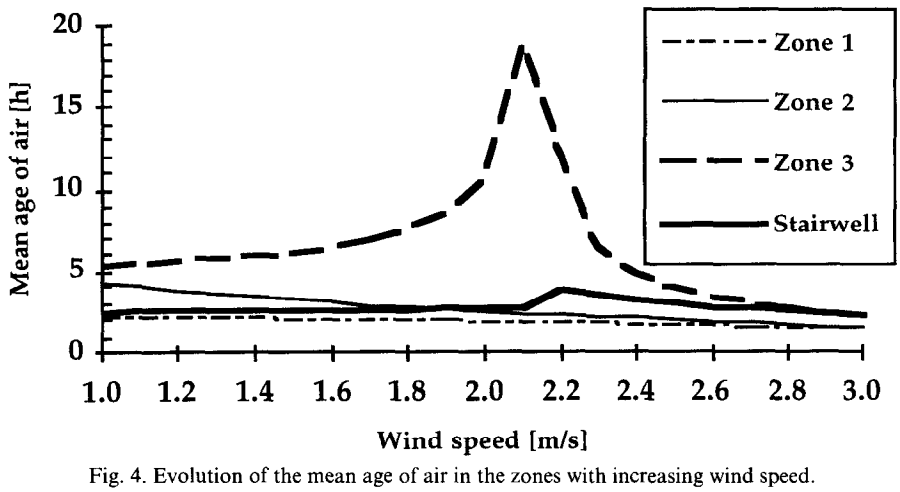


Fig. 4. Evolution of the mean age of air in the zones with increasing wind speed.

first order interaction coefficients are aliased between themselves. A description of the aliases of this design is given in [2].

Sensitivity and effects

The variation of the age of air when the wind is increasing is shown in Fig. 4. The third zone is in a critical situation when the stack pressure compensates the wind pressure near 2.1 m s^{-1}. The fresh air cannot enter from the window and very little air enters from the stairwell. The consequence is a very high age of air in this zone.

A similar phenomenon occurs for the stairwell near a wind speed of 2.2 m s^{-1}. The equilibrium between stack and wind pressure at the low opening of this zone results in less ventilation than if either force dominates. (When the stack effect dominates, fresh air enters, whereas it escapes when the wind is dominant, cf. Table 4.)

The other two zones show monotonic behaviour, i.e. the mean age of air decreases when the wind increases.

The air flows with the corresponding flow matrices are shown in Table 4. The elements of the flow matrix are defined as follows:

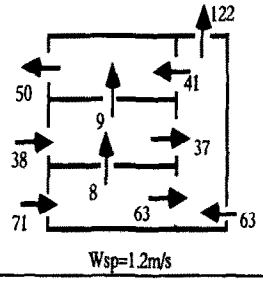
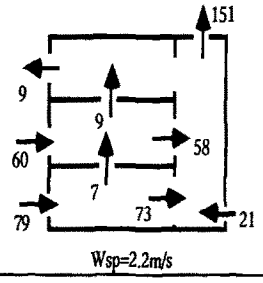
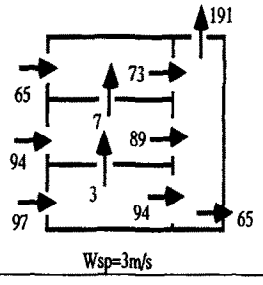
$$Q_{ij} = -q_{ij}$$

$$Q_{ii} = \sum_{j=0}^{N} q_{ij}(1-\delta_{ij}). \tag{10}$$

The sensitivity coefficients for 1% of variation are given in the Appendix with some statistical indicators. Except around the critical situation described above, the standard deviation does not vary significantly with the wind speed variation. Figure 5 shows the standard deviation for the mean age of air for all four zones. As in other cases, in a critical situation, when the flows are very weak, one can observe extreme sensitivity.

Figure 6 presents the evolution of the largest 'main effects' of the global mean age with the increase of wind speed. The critical situation appears clearly, even hiding

Table 4. Flow matrices and flows for typical Archimedes numbers

Wind speed	Ar	Flow matrix	Flow scheme
1.3m/s	2.5	$\begin{array}{c} l/s \\ -71 \\ -38 \\ 0 \\ -63 \end{array} \begin{pmatrix} 0 & & -50 & -122 \\ 71 & 0 & 0 & 0 \\ -8 & 46 & 0 & 0 \\ 0 & -9 & 50 & -41 \\ -63 & -37 & 0 & 163 \end{pmatrix}$	
2.2m/s	0.7	$\begin{array}{c} l/s \\ -79 \\ -60 \\ 0 \\ -21 \end{array} \begin{pmatrix} 0 & & -9 & -151 \\ 79 & 0 & 0 & 0 \\ -7 & 67 & 0 & 0 \\ 0 & -9 & 9 & 0 \\ -73 & -58 & 0 & 151 \end{pmatrix}$	
3m/s	0.4	$\begin{array}{c} l/s \\ -97 \\ -94 \\ -65 \\ 0 \end{array} \begin{pmatrix} 0 & 0 & 0 & -256 \\ 97 & 0 & 0 & 0 \\ -3 & 97 & 0 & 0 \\ 0 & -7 & 73 & 0 \\ -94 & -89 & -73 & 256 \end{pmatrix}$	

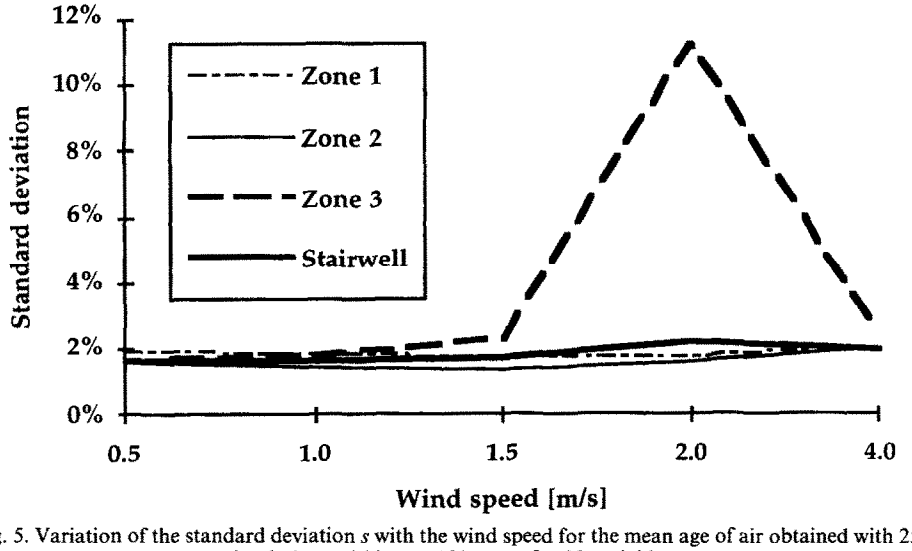


Fig. 5. Variation of the standard deviation s with the wind speed for the mean age of air obtained with 256 simulations within a $\pm 1\%$ range for 23 variables.

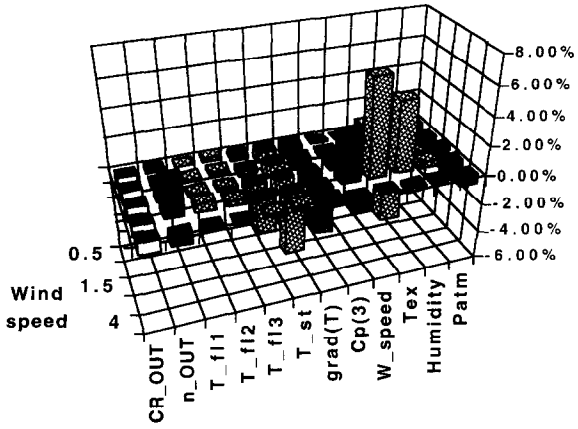


Fig. 6. Evolution of the largest 'main effects' with the wind speed.

the monotonic evolution of the effects of wind speed and temperatures.

Comparing a stack dominated situation with a wind dominated one, the following can be said for Fig. 7.

- The wind speed (W_speed) effect increases with the wind speed, but the relation is not linear as expected.
- The same can be observed for the pressure coefficients (Cp(i)).
- The inverse is observed for temperatures (Tex, T_fl2, T_fl3, T_st) and the temperature gradient in the stair well (grad(T)) whose effects decrease when the wind speed increases.
- The other dominant parameters are the outdoor–indoor airtightness (CR_OUT) and the atmospheric pressure (1% variation in Patm corresponds to about 300 m height or weather change).
- The effect of the exponent (n_OUT) becomes significant when the wind dominates the infiltration.

Let us now compare the results for a different input variation amplitude. In Fig. 8, the mean values of the age of air corresponding to the constant effect α_0 are plotted as a function of the amplitude. The 0% values are obtained when making a simulation at the centre of the domain. So, when the values differ from the central result, the model is non-linear because the interpolated value does not fit the calculated one.

In Fig. 8, the behaviour is linear up to 5%; after that, the difference between the interpolated and the calculated values indicates a curvature of the surface response.

In Fig. 9, the effects of the input variation amplitude are evaluated at the level of the global sensitivity. For the age of air in each zone, the RMS of the main effects and the interaction effects have been normalised by the amplitude. It is for the smallest amplitude, 1%, that the sensitivity is at its maximum. This can be explained by the curvature of the surface response. The simulation being performed at the vertex of the domain, a possible elevation at the centre does not affect the effects. Finally,

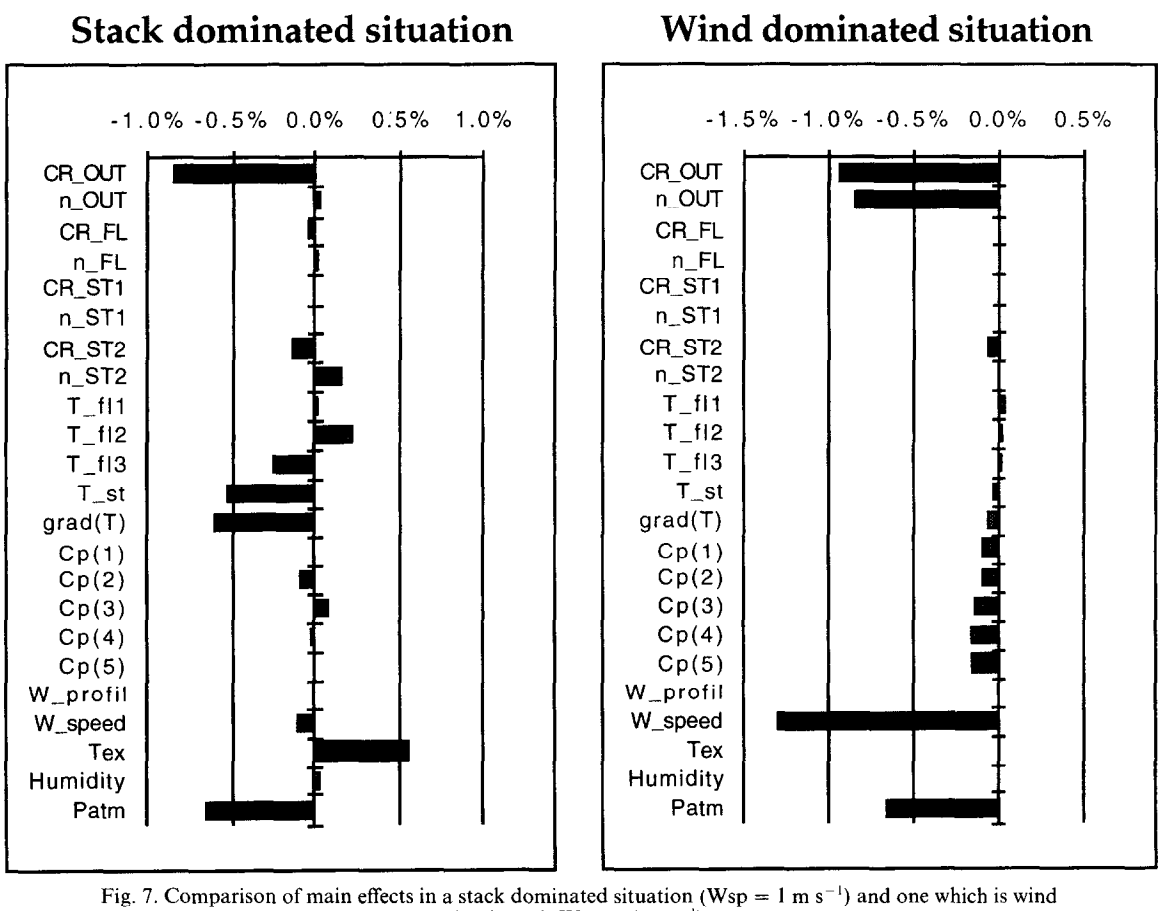


Fig. 7. Comparison of main effects in a stack dominated situation (Wsp = 1 m s^{-1}) and one which is wind dominated (Wsp = 4 m s^{-1}).

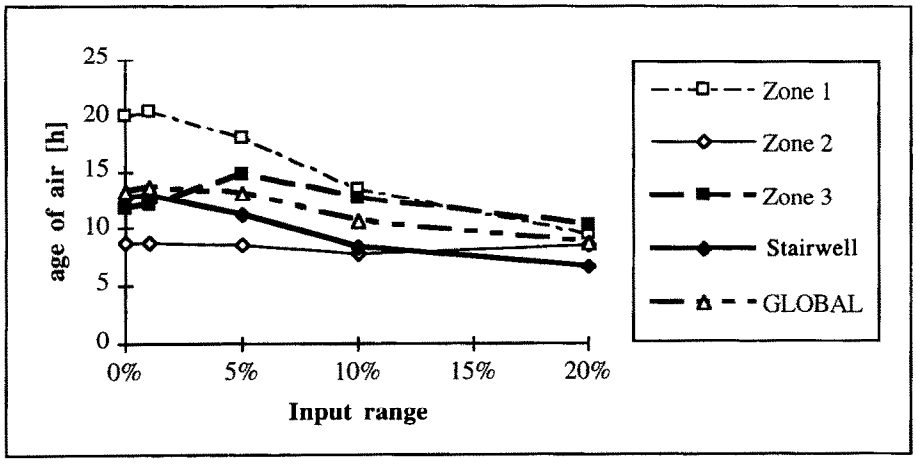


Fig. 8. Evolution of the mean value with the increase of the input variation amplitude (wind : 2 m s^{-1}).

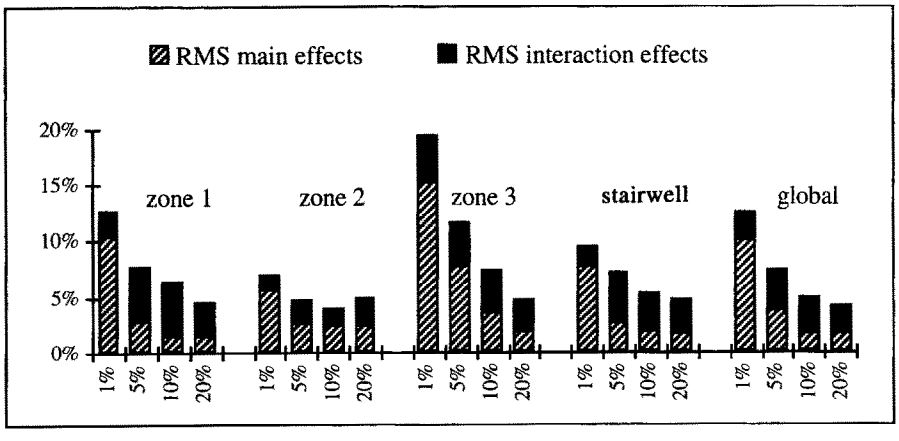


Fig. 9. Comparison of the standard deviation due to the main effects and due to the interaction effects for all zones and for four amplitudes. The values are standardised by dividing the RMS values by the magnitude of the range.

the sum of the interaction effects is similar for all the amplitudes.

TREATMENT WITH MONTE-CARLO METHOD

The analysis of Monte-Carlo simulations is straightforward in comparison with the factorial method: the mean $m(Y_i)$ and standard deviation $s(Y_i)$ are calculated for each selected output parameter Y_i. Nevertheless, a check of the distribution of the output and of the convergence of the mean and the standard deviation gives some confidence in the results.

The histograms of Fig. 10 have been obtained using a Gaussian distribution with a standard deviation of 1% for the variation of the 23 selected input parameters (cf. Table 3). These histograms of 200 runs show quasi-normal distributions, but all the output series have failed the χ^2 test for normality at 10% probability. The usual assumption that the behaviour of everything can be described by Gaussian distribution is then to be considered with caution since the calculation of the confidence interval is based on the Gaussian hypothesis.

Figure 11 presents the convergence of the standard deviation $s(Y_i)$ when the number of runs increases. For almost 80 runs, the RMS is in the vicinity of its confidence intervals. The same type of behaviour can be observed with the mean $m(Y_i)$. If the distribution of the input parameters is changed for a uniform distribution with also 1% standard deviation, the same mean values and the same deviations $s(Y_i)$ are obtained as long as the response surface is linear (Table 5). In the case of curvature, the larger number of points at the centre of the domain for the Gaussian distribution influences the mean values and the deviation (Table 6).

Figures 12 and 13 correspond to Figs 4 and 5. They show the variations of the mean value $m(Y_i)$ and the standard deviation $s(Y_i)$ of the age of air in the four zones when the Archimedes number varies around unity. The same behaviour with the same magnitude is observed, showing the same maximum when the wind and the stack effect are in concurrence at 2 m s^{-1}.

COMPARISON OF THE TWO METHODS

From what is presented above, the reader can see the differences between the two methods, which are summarised in Table 7. However, more than the differences at the level of the operating features, it seems important to compare the sensitivity assessed by one or the other method. To do that, a numerical test was performed

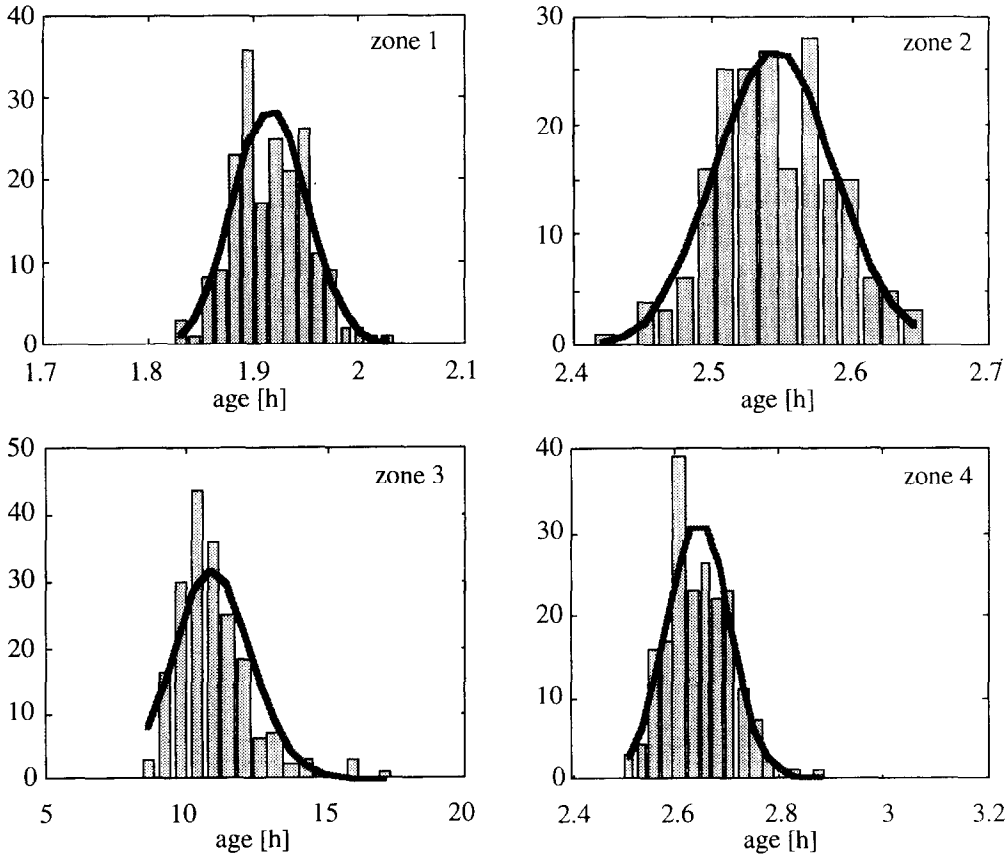


Fig. 10. Histograms of the age of air of the zones for 200 runs. The 23 input parameters have been varied randomly following a normal distribution with a standard deviation of 1% of the mean value of each selected input parameter.

which compared the sensitivity obtained with both methods for randomly chosen surface responses. The structure of the test is presented in Fig. 14. The results in Fig. 15 demonstrate that, statistically, the sensitivity determined by both methods is the same. However, depending on the actual shape of the response surface, a gap can exist between the results of the two methods. It is also evident that if the response surface has a curvature, the Monte-Carlo method with a Gaussian distribution favours the centre of the domain when factorial design cuts the influence of a central extremum which is detected only by an additional run at the centre of the domain. In cases of curvature of the domain, another design with square terms should be preferred.

COMBINATION

Before concluding this panegyric of the sensitivity analysis, we would like to illustrate that both methods presented here can be used synergetically. This synergy works particularly well for quality assessment; one could be interested not only in estimating the sensitivity in a domain but also in finding where in a domain the sensitivity is larger or smaller, i.e. one could be interested in drawing a map of the sensitivity over a domain. To do that, the factorial design can be used to visit the domain and the Monte-Carlo method to estimate the sensitivity at each point of the factorial design. The ranges for both designs must have a ratio of one order of magnitude. This will correspond to fitting a linear function, not on the response Y [cf. equation (1)], but on its local standard deviation S_y.

$$S_y = \gamma_0 + \sum_i \gamma_i X_i + \sum_{i \neq j} \gamma_{ij} X_i X_j + \dots \qquad (11)$$

The experimental matrix will then have the following aspect for three dimensions:

$$\underbrace{\begin{bmatrix} -1 & -1 & -1 \\ 1 & -1 & -1 \\ -1 & 1 & -1 \\ -1 & -1 & 1 \\ 1 & 1 & -1 \\ 1 & -1 & 1 \\ -1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}}_{\text{factorial design around } (0,0,0)} \rightarrow \underbrace{\begin{bmatrix} x_{11} & x_{12} & x_{13} \\ \vdots & \ddots & \vdots \\ x_{m1} & x_{m2} & x_{m3} \\ & \vdots & \\ & \vdots & \\ & \vdots & \\ & \vdots & \\ & \vdots & \\ & \vdots & \\ & \vdots & \\ & \vdots & \\ & \vdots & \end{bmatrix}}_{\text{combined design}} \left.\begin{matrix} \\ \\ \\ \end{matrix}\right\} \begin{matrix} \text{MC} \\ \text{design} \\ \text{around} \\ (-1,-1,-1) \end{matrix}$$

This technique was used for the simulation of the test building studied above. The sensitivity of the age of air in the four zones was investigated for the domain determined by two parameters involved in the Archimedes number: the wind speed v and the indoor–outdoor temperature difference ΔT. At different locations of this

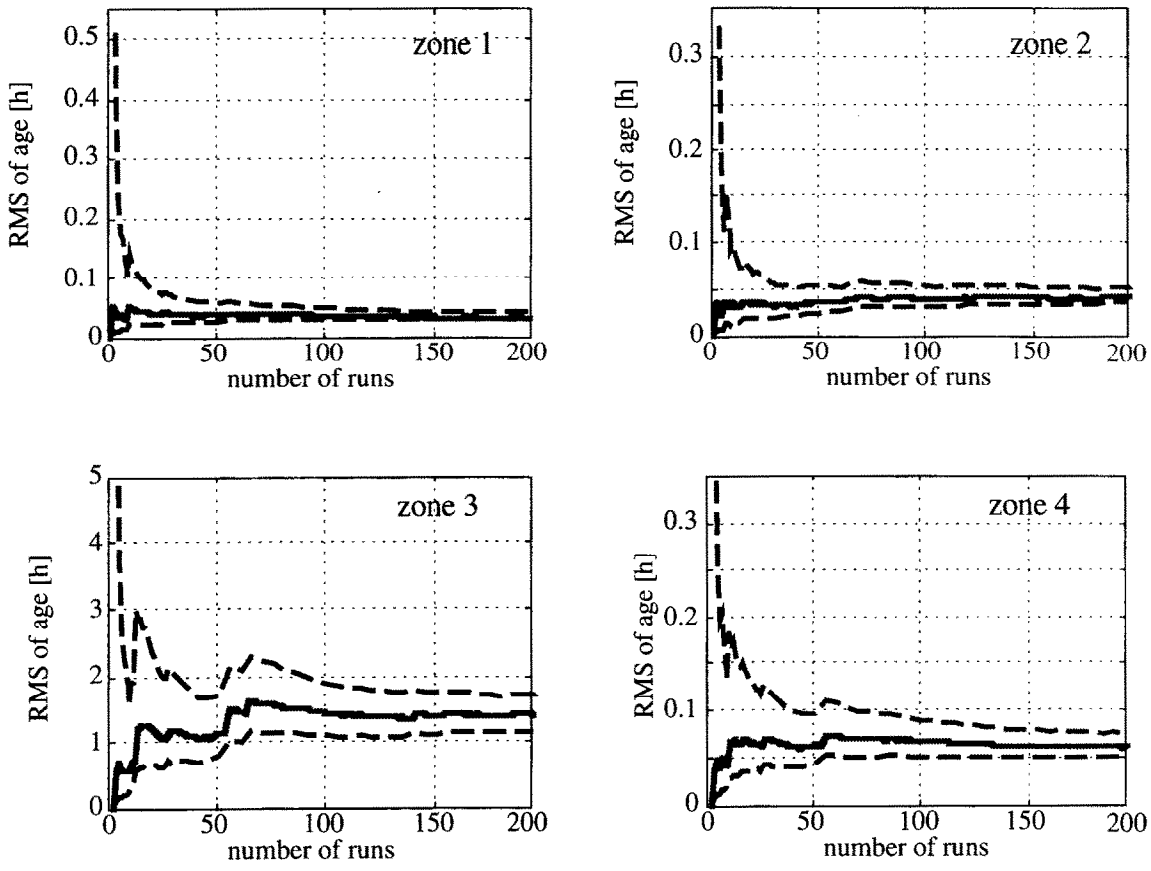


Fig. 11. Convergence in the Monte-Carlo method of the RMS of the age of air in the four zones of the test building when using Gaussian distribution with a standard deviation of 1% for the random values of the input parameters.

Table 5. Comparison between mean values and standard deviation of the age of air when using Monte-Carlo method with Gaussian or uniform distribution with a standard deviation of 1% for the random values of the input parameters

	Distribution	Zone 1	Zone 2	Zone 3	Zone 4
Mean	$N(0,1)$	1.91	2.54	11.00	2.64
	$U(-\sqrt{3},\sqrt{3})$	1.91	2.54	10.98	2.64
RMS	$N(0,1)$	0.036	0.044	1.40	0.061
	$U(-\sqrt{3},\sqrt{3})$	0.036	0.038	1.28	0.060

Table 6. Comparison between mean values and standard deviation of the age of air when using Gaussian or uniform distribution with a standard deviation of 20% for the random values of the input parameters

	Distribution	Zone 1	Zone 2	Zone 3	Zone 4
Mean	$N(0,1)$	1.98	2.65	9.23	2.98
	$U(-\sqrt{3},\sqrt{3})$	1.97	2.58	9.65	2.99
RMS	$N(0,1)$	0.37	0.48	5.31	0.79
	$U(-\sqrt{3},\sqrt{3})$	0.38	0.41	8.54	0.79

domain, a Monte-Carlo analysis was performed involving 23 parameters. The locations in the domain ($\Delta T \times v$) were chosen according to the factorial design. So, considering as response the standard deviation resulting from the Monte-Carlo simulation, it was possible to analyse the sensitivity of the standard deviation within the domain. This technique requires two types of variation amplitude: the larger one to move within the domain

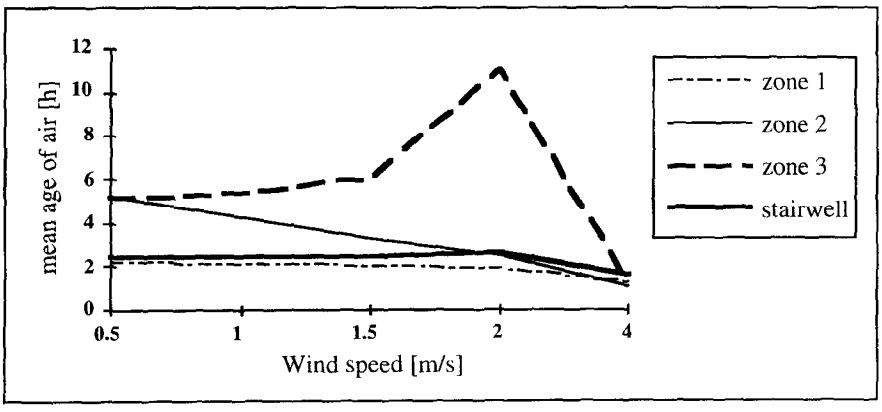


Fig. 12. Average of the mean age of air in the zones as function of the increase of the wind speed when using the Monte-Carlo method with normal distribution with a standard deviation of $\pm 1\%$ for 23 inputs.

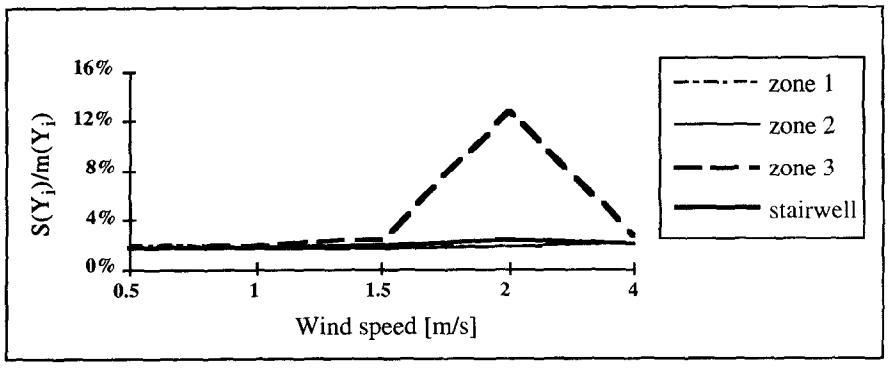


Fig. 13. Standard deviation $s(Y_i)$ as function of the wind speed for the mean age of air obtained with 256 simulations with a Monte-Carlo method using a normal distribution with a standard deviation of $\pm 1\%$ for 23 inputs.

Table 7. Comparison of the operating features of the Monte-Carlo and the factorial design method

Features	Monte-Carlo	Factorial design
Number of runs	less than 100 is theoretically sufficient for any number of input parameters	fraction of 2^N if N is the number of input parameters
Parameter number	unimportant	depends on available design
Ranges	according to the probability distribution of each input parameter	minima and maxima of each input parameter
Final information	global sensitivity	global and individual sensitivities
Analysis calculation	mean value, standard deviation and histogram	effects by matrix product and inversion
	$m_Y = \frac{1}{N_{\text{sim}}} \sum_{k=1}^{N_{\text{sim}}} Y_k$	$\alpha = (X^T X)^{-1} X^T Y$
	$(s_Y)^2 = \frac{1}{N_{\text{sim}}} \sum_{k=1}^{N_{\text{sim}}} (Y_k - m_Y)^2$	

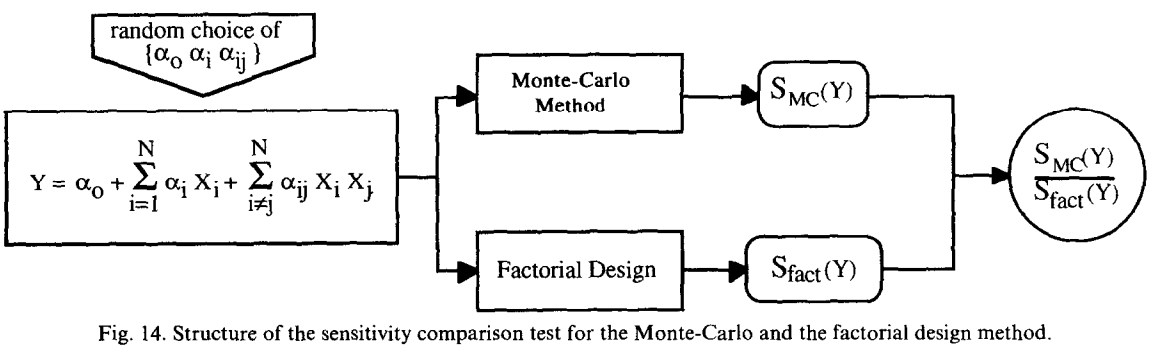


Fig. 14. Structure of the sensitivity comparison test for the Monte-Carlo and the factorial design method.

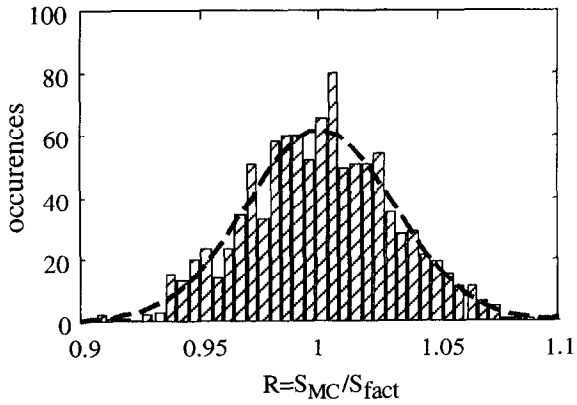


Fig. 15. Histogram for 1000 runs of the ratio between the standard deviation S_{MC} obtained with the Monte-Carlo method and S_{fact} obtained with a full factorial design. The tested models have 10 parameters with first order interactions. The coefficients are chosen at random at each run.

($\Delta T \times v$) and the smaller one, to define the vicinity of each investigated point. Figure 16 shows the fit of these responses on a linear function and Fig. 17 a square function.

The number and the position of the points depends on the degree of the fitted function. For the linear function, the factorial design is optimal. For the second degree, it is necessary to add points at the centre and at the middle of the ridge of the domain: this design is called hybrid.

The following comments have to be made regarding Figs 16 and 17.

- There is not a large difference in behaviour between the one-degree and two-degree models. Both show the same type of behaviour. Only the behaviour of the standard deviation of zone 3 appears slightly different when fitted on one function or the other.
- In zone 1, the standard deviation increases with the outdoor temperature and decreases with the wind speed. Its sensitivity then does not depend on the Archimedes number whose iso-curves are presented in Fig. 18.
- In zone 2, the standard deviation in sensitive only to the wind speed.
- In zone 3, the behaviour is much more difficult to analyse.
- In zone 4, the behaviour of the standard deviation could be related to the Archimedes number since the isolines are more or less parallel.

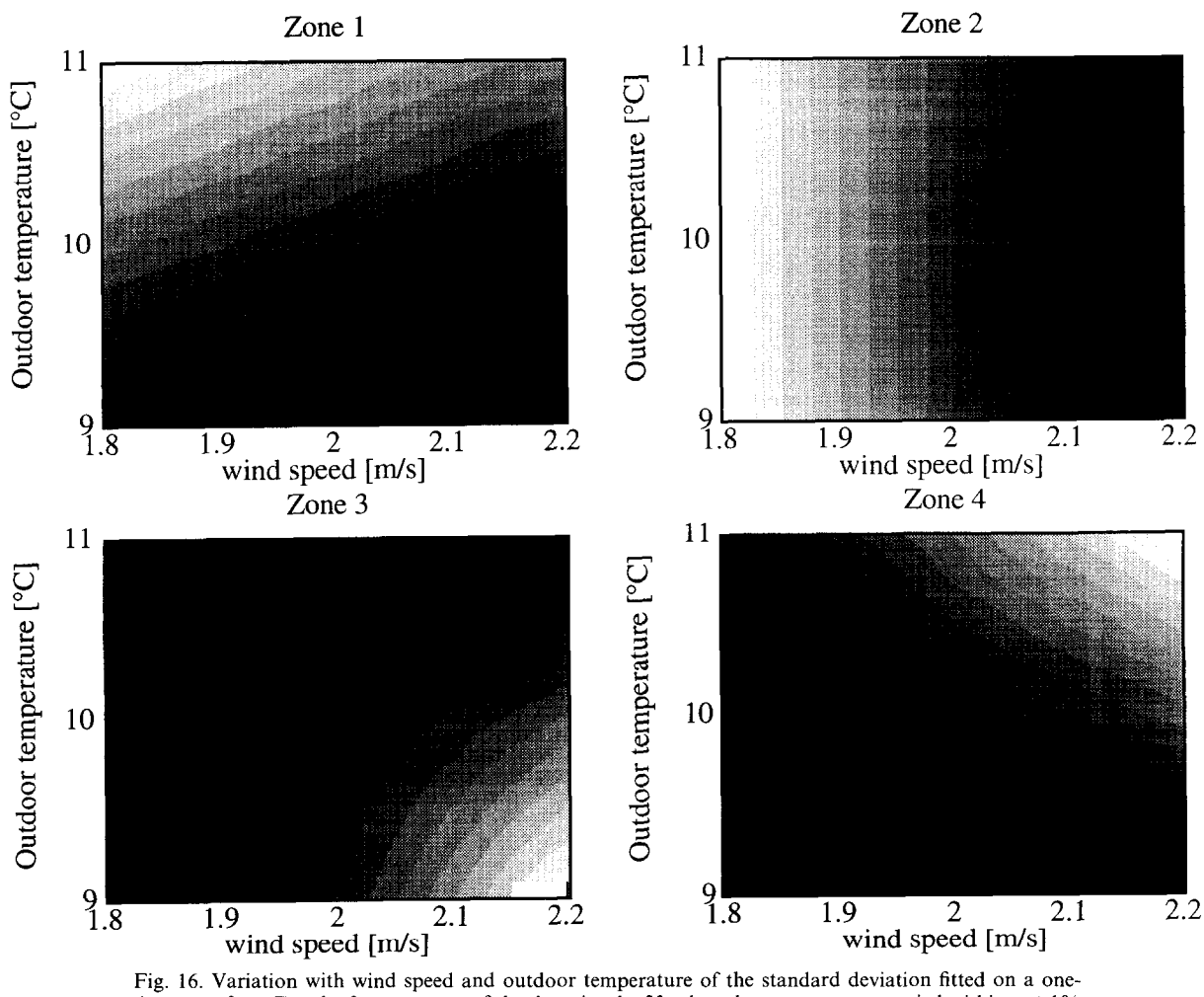


Fig. 16. Variation with wind speed and outdoor temperature of the standard deviation fitted on a one-degree surface. For the four vertexes of the domain, the 23 selected parameters are varied within a $\pm 1\%$ range using the Monte-Carlo method. The white zone corresponds to the maximum.

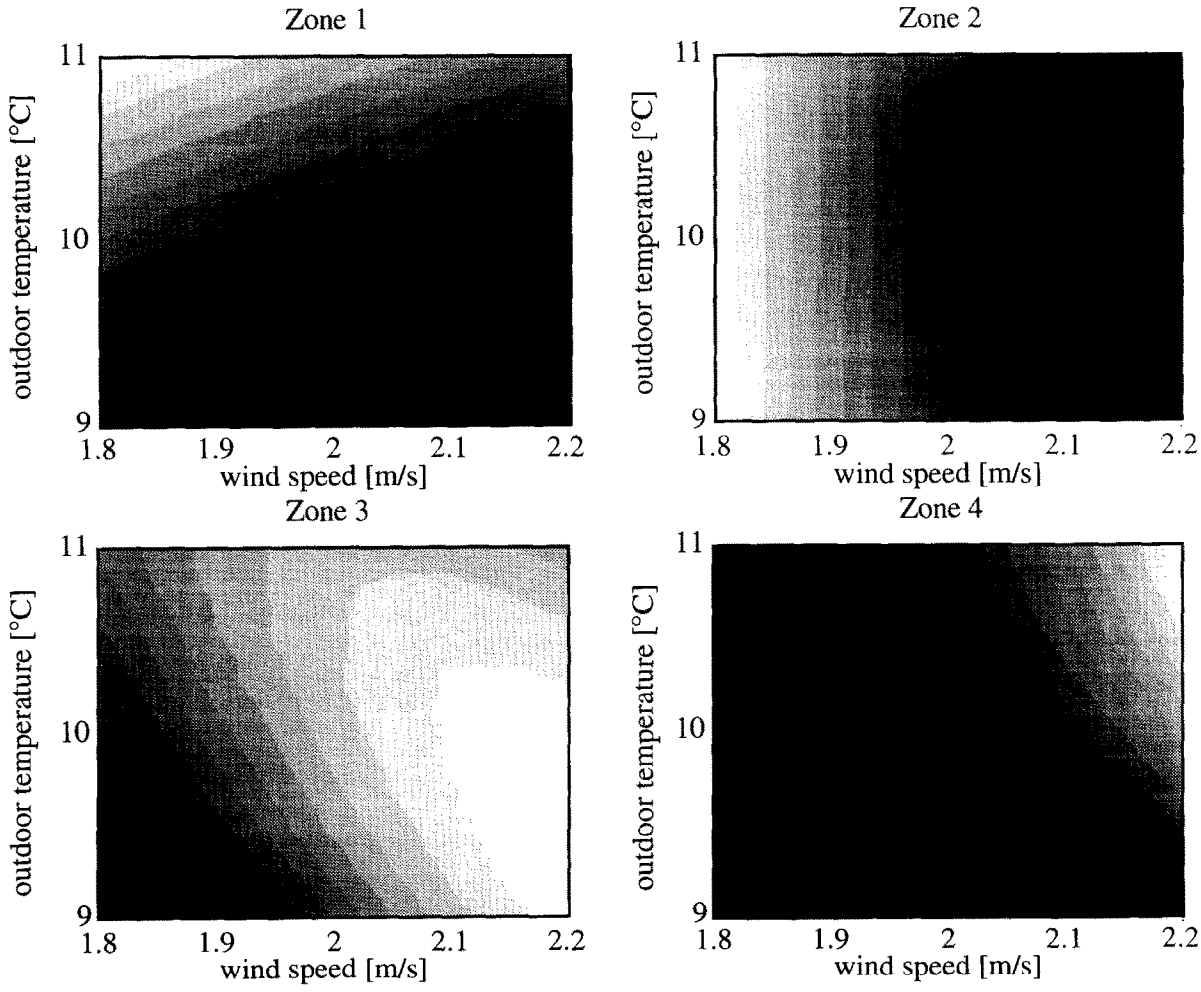


Fig. 17. Variation with the wind speed and the outdoor temperature of the standard deviation fitted on a second-degree surface. For nine points of the domain, the 23 selected parameters are varied within a $\pm 1\%$ range using the Monte-Carlo method. The white zone corresponds to the maximum.

CONCLUSION

The use of factorial design in simulation when investigating the sensitivity of a simulation program has been illustrated. Comparison with the commonly used Monte-Carlo method allowed highlighting of the advantages of the factorial design which yields more information for an equivalent amount of work.

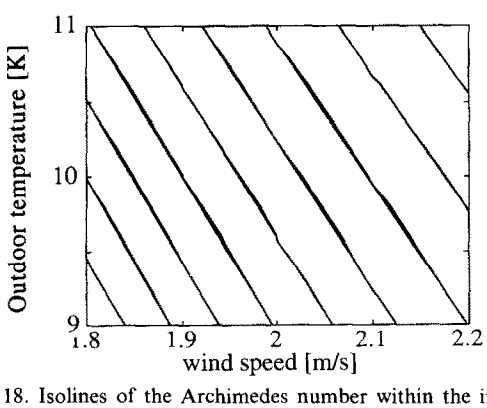


Fig. 18. Isolines of the Archimedes number within the investigated domain.

A large number of runs can be saved by using fractional factorial design. Although the literature abounds, there are very few computerised tools for the design of experiments. The procedures to select the optimal design should be made more user friendly to be of general use.

The test case has demonstrated that the interaction between wind and stack effect is not straightforward and a simple structure can present a relatively complex behaviour.

The integration in simulation tools of a sensitivity analysis module is the right way to produce robust simulation tools. This will allow the user to determine the consequence of the inaccuracies of the input and to compute the confidence intervals of output data. Such features are under development within Annex 23 'Multizone Air Flow Modelling' of the IEA ECB&C program.

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APPENDIX

Table A1. Effects of the age of air in the four zones of the test building for the variation of $\pm 1\%$ of the 23 input parameters. The mean wind speed is 0.5 m s^{-1}. Values are percentages

		Zone 1	Zone 2	Zone 3	Stairwell	GLOBAL
Constant effect	[h]	2.2	5.3	5.2	2.4	3.8
Response in the centre	[h]	2.2	5.3	5.2	2.4	3.8
CR_OUT		−0.97	−0.59	−0.85	−0.94	−0.79
n_OUT		−0.09	0.19	0.09	−0.13	0.06
CR_FL		−0.01	−0.18	0.02	−0.05	−0.07
n_FL		0.00	0.09	−0.01	0.02	0.03
CR_ST1		0.00	0.00	0.00	0.00	0.00
n_ST1		0.00	0.00	0.00	0.00	0.00
CR_ST2		−0.02	−0.23	−0.16	−0.01	−0.14
n_ST2		0.01	0.36	0.19	−0.01	0.20
T_fl1		0.46	−0.52	0.05	0.15	−0.08
T_fl2		−0.17	0.88	0.25	−0.15	0.36
T_fl3		−0.09	−0.35	−0.26	−0.18	−0.25
T_st		−0.93	0.10	−0.70	−0.68	−0.44
grad(T)		−0.72	−0.64	−0.68	−0.55	−0.65
Cp(1)		−0.02	0.01	0.00	0.00	0.00
Cp(2)		0.01	−0.09	−0.02	0.00	−0.03
Cp(3)		0.01	0.03	0.03	0.01	0.02
Cp(4)		−0.01	−0.02	0.00	−0.01	−0.01
Cp(5)		−0.01	−0.02	0.00	0.01	−0.01
W_profil		0.00	0.00	0.00	0.00	0.00
W_speed		−0.02	−0.18	0.01	0.01	−0.06
Tex		0.87	−0.07	0.73	0.81	0.48
Humidity		0.06	0.00	0.05	0.06	0.03
Patm		−0.65	−0.65	−0.65	−0.65	−0.65
RMS main effects		2	2	2	2	1
RMS interaction		0.04	0.03	0.03	0.03	0.03
RMS all effects		2	2	2	2	1
RMS all runs		2	2	2	2	1

Table A2. Effects of the age of air in the four zones of the test building for the variation of ±1% of the 23 input parameters. The mean wind speed is 1 m s^{-1}. Values are percentages

		Zone 1	Zone 2	Zone 3	Stairwell	GLOBAL
Constant effect	[h]	2.2	4.3	5.4	2.4	3.6
Response in the centre	[h]	2.2	4.3	5.4	2.4	3.6
CR_OUT		−0.98	−0.67	−0.85	−0.94	−0.83
n_OUT		−0.17	0.09	0.17	−0.13	0.05
CR_FL		−0.01	−0.12	0.01	−0.04	−0.04
n_FL		0.00	0.06	0.00	0.02	0.02
CR_ST1		0.00	0.00	0.00	0.00	0.00
n_ST1		0.00	0.00	0.00	0.00	0.00
CR_ST2		−0.01	−0.21	−0.15	−0.02	−0.13
n_ST2		0.00	0.31	0.19	0.02	0.17
T_fl1		0.43	−0.33	0.08	0.12	0.02
T_fl2		−0.14	0.63	0.20	−0.06	0.24
T_fl3		−0.10	−0.27	−0.31	−0.17	−0.24
T_st		−0.88	0.06	−0.77	−0.70	−0.52
grad(T)		−0.69	−0.46	−0.68	−0.60	−0.60
Cp(1)		−0.07	0.03	−0.01	−0.02	−0.01
Cp(2)		0.04	−0.26	−0.06	−0.01	−0.09
Cp(3)		0.04	0.09	0.13	0.05	0.09
Cp(4)		−0.03	−0.06	0.00	−0.03	−0.03
Cp(5)		−0.03	−0.05	0.02	0.02	−0.01
W_profil		0.00	0.00	0.00	0.00	0.00
W_speed		−0.09	−0.51	0.14	0.03	−0.11
Tex		0.82	−0.04	0.83	0.80	0.56
Humidity		0.06	0.00	0.06	0.05	0.04
Patm		−0.65	−0.65	−0.65	−0.65	−0.65
RMS main effects		2	1	2	2	2
RMS interaction		0.04	0.03	0.04	0.03	0.03
RMS all effects		2	1	2	2	2
RMS all runs		2	1	2	2	2

Table A3. Effects of the age of air in the four zones of the test building for the variation of ±1% of the 23 input parameters. The mean wind speed is 1.5 m s^{-1}. Values are percentages

		Zone 1	Zone 2	Zone 3	Stairwell	GLOBAL
Constant effect	[h]	2.1	3.3	6.1	2.5	3.5
Response in the centre	[h]	2.1	3.3	6.1	2.5	3.5
CR_OUT		−1.00	−0.74	−0.84	−0.95	−0.86
n_OUT		−0.27	−0.04	0.44	−0.16	0.12
CR_FL		−0.01	−0.07	−0.01	−0.02	−0.03
n_FL		0.00	0.04	0.00	0.01	0.01
CR_ST1		0.00	0.00	0.00	0.00	0.00
n_ST1		0.00	0.00	0.00	0.00	0.00
CR_ST2		0.01	−0.19	−0.15	−0.03	−0.11
n_ST2		0.00	0.26	0.21	0.04	0.16
T_fl1		0.40	−0.18	0.13	0.11	0.10
T_fl2		−0.11	0.45	0.16	0.00	0.16
T_fl3		−0.10	−0.21	−0.48	−0.17	−0.31
T_st		−0.80	0.03	−1.02	−0.75	−0.69
grad(T)		−0.63	−0.31	−0.83	−0.66	−0.65
Cp(1)		−0.14	0.04	−0.04	−0.03	−0.04
Cp(2)		0.07	−0.40	−0.13	−0.03	−0.15
Cp(3)		0.09	0.14	0.41	0.13	0.25
Cp(4)		−0.05	−0.08	0.02	−0.07	−0.03
Cp(5)		−0.06	−0.10	0.07	0.06	0.01
W_profil		0.00	0.00	0.00	0.00	0.00
W_speed		−0.13	−0.53	0.45	0.06	0.06
Tex		0.75	−0.02	1.17	0.83	0.76
Humidity		0.05	0.00	0.08	0.06	0.05
Patm		−0.65	−0.66	−0.65	−0.65	−0.65
RMS main effects		2	1	2	2	2
RMS interaction		0.0	0.0	0.1	0.0	0.1
RMS all effects		2	1	2	2	2
RMS all runs		2	1	2	2	2

Table A4. Effects of the age of air in the four zones of the test building for the variation of ±1% of the 23 input parameters. The mean wind speed is 2 m s^{-1}. Values are percentages

		Zone 1	Zone 2	Zone 3	Stairwell	GLOBAL
Constant effect	[h]	1.9	2.5	11.0	2.6	4.6
Response in the centre	[h]	1.9	2.5	10.8	2.6	4.5
CR_OUT		−1.02	−0.78	−0.81	−0.97	−0.85
n_OUT		−0.34	−0.14	2.43	−0.25	1.41
CR_FL		0.00	−0.04	−0.08	0.00	−0.06
n_FL		0.00	0.02	0.01	0.00	0.01
CR_ST1		0.00	0.00	0.00	0.00	0.00
n_ST1		0.00	0.00	0.00	0.00	0.00
CR_ST2		0.02	−0.18	−0.21	−0.03	−0.16
n_ST2		−0.02	0.19	0.32	0.04	0.23
T_fl1		0.38	−0.07	0.53	0.11	0.37
T_fl2		−0.07	0.35	0.34	0.05	0.26
T_fl3		−0.13	−0.19	−2.28	−0.24	−1.48
T_st		−0.74	−0.03	−4.19	−1.00	−2.80
grad(T)		−0.58	−0.24	−3.12	−0.86	−2.14
Cp(1)		−0.24	0.03	−0.36	−0.06	−0.25
Cp(2)		0.09	−0.53	−0.37	−0.08	−0.30
Cp(3)		0.21	0.25	3.14	0.36	2.04
Cp(4)		−0.07	−0.08	0.48	−0.12	0.26
Cp(5)		−0.12	−0.14	0.78	0.16	0.47
W_profil		0.00	0.00	0.11	0.00	0.07
W_speed		−0.28	−0.93	6.61	0.51	3.97
Tex		0.69	0.01	4.89	1.09	3.23
Humidity		0.05	0.00	0.39	0.07	0.26
Patm		−0.65	−0.65	−0.65	−0.65	−0.65
RMS main effects		2	2	11	2	7
RMS interaction		0.0	0.0	2.7	0.1	1.7
RMS all effects		2	2	11	2	7
RMS all runs		2	2	11	2	7

Table A5. Effects of the age of air in the four zones of the test building for the variation of ±1% of the 23 input parameters. The mean wind speed is 4 m s^{-1}. Values are percentages

		Zone 1	Zone 2	Zone 3	Stairwell	GLOBAL
Constant effect	[h]	1.3	1.1	1.2	1.6	1.3
Response in the centre	[h]	1.3	1.1	1.2	1.6	1.3
CR_OUT		−1.13	−0.86	−0.83	−0.93	−0.94
n_OUT		−0.98	−0.81	−0.73	−0.84	−0.84
CR_FL		0.00	−0.01	0.01	0.00	0.00
n_FL		0.00	0.01	−0.01	0.00	0.00
CR_ST1		0.00	0.00	0.00	0.00	0.00
n_ST1		0.00	0.00	0.00	0.00	0.00
CR_ST2		0.13	−0.13	−0.18	−0.07	−0.06
n_ST2		−0.02	0.02	0.02	0.01	0.01
T_fl1		0.19	−0.01	−0.05	0.04	0.05
T_fl2		−0.03	0.11	0.00	0.03	0.02
T_fl3		−0.04	−0.05	0.15	0.02	0.02
T_st		−0.36	0.01	0.25	−0.03	−0.04
grad(T)		−0.29	−0.06	0.18	−0.06	−0.06
Cp(1)		−0.44	0.09	0.10	−0.08	−0.09
Cp(2)		0.17	−0.66	0.14	−0.11	−0.10
Cp(3)		0.27	0.20	−0.86	−0.14	−0.14
Cp(4)		−0.21	−0.13	−0.14	−0.16	−0.16
Cp(5)		−0.21	−0.14	−0.15	−0.17	−0.17
W_profil		0.00	0.00	0.00	0.00	0.00
W_speed		−0.83	−1.26	−1.83	−1.31	−1.30
Tex		0.34	−0.01	−0.28	0.01	0.02
Humidity		0.02	0.00	−0.02	0.00	0.00
Patm		−0.65	−0.65	−0.65	−0.65	−0.65
RMS main effects		2	2	2	2	2
RMS interaction		0.0	0.0	0.0	0.0	0.0
RMS all effects		2	2	2	2	2
RMS all runs		2	2	2	2	2